

Modeling Iowa's Historic-to-Present Vehicular Crash Data to Improve Public Safety

MIS 5460 – Spring 2025
Final Project Report

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1 INTRODUCTION

1.1 BACKGROUND

Motor vehicle crashes are still a major concern regarding public safety, transport networks, and mobility projects within the United States. Each year, thousands of crashes in Iowa stem from weather, traffic splits, and rising car counts. The availability of thorough crash datasets has opened new opportunities for data-driven approaches for understanding as well as reducing traffic incidents. Through the merging of advanced data analytics, geographic mapping, and interactive visualization tools, such as Tableau, researchers and policymakers can detect trends, identify root causes, and propose targeted safety interventions more effectively than previously.

Through this project, a strong analysis of Iowa's vehicular crash data is aimed to be provided. Statistical methods, visual tools, and predictive models are planned to be leveraged in this effort. By presenting insights in an interactive and accessible manner, the goal is to support initiatives aimed at improving road safety, reducing economic costs, and protecting lives across the state.

1.2 OBJECTIVE

This project aims to conduct a data-driven analysis of vehicle crash data in Iowa from January 2009 to March 2025. The goal is to uncover the factors that most significantly contribute to traffic accidents and identify patterns over time and geography. The insights gained from this analysis can support evidence-based decisions for improving road safety and traffic regulations in Iowa.

1.3 SCOPE AND GOALS

The key goals of the project include:

- Creating interactive dashboards in Tableau to visualize crash patterns across time, location, and severity.
- Modeling relationships between crash features and outcomes (such as severity).
- Identifying areas prone to crashes based on historical data
- Presenting actionable insights and recommendations through visualizations and a final report.

Deliverables include:

- Geographic visualization of crash data
- Tableau visualizations of geographic, temporal, and categorical crash analysis
- Predictive model showing how various factors influence the likelihood of property damage, injury, or fatality

1.4 DATASET DESCRIPTION

The dataset contains traffic crash records from January 2009 to March 2025 in the state of Iowa (Iowa Department of Transportation, Motor Vehicle Division). It includes various attributes such as:

- Date and time of the crash
- Location coordinates or descriptions
- Road conditions and weather
- Involved vehicle types
- Lighting and environmental factors
- Crash severity and contributing circumstances

This dataset offers a comprehensive basis for analyzing trends, identifying high-risk scenarios, and supporting data-informed traffic safety initiatives.

2 EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was conducted to better understand the structure, distribution, and relationships within the Iowa crash dataset. This step aimed to identify patterns, detect anomalies, and guide subsequent modeling efforts.

2.1 PREPROCESSING

Before analysis, the dataset underwent cleaning to handle:

- **Missing values:** Several columns had missing data, ranging from minimal gaps (~0.01%) to extremely high percentages (up to 98%). Variables such as *Work Zone* and *Travel Direction* had the highest missingness and were handled by imputing with placeholders like "Unknown" or excluding them from modeling.
- **Categorical data handling:** For categorical features, missing values were filled with consistent placeholders such as "Not Provided" to avoid skewing analysis. Variables like *weather*, *lighting conditions*, and *surface conditions* were later encoded numerically using `LabelEncoder`.
- **Numerical imputation:** Missing numerical data was imputed using statistical methods—mean or median—depending on the distribution and context. In some cases, zero imputation was applied where semantically meaningful.
- **Feature scaling:** To prepare for modeling, numeric features were normalized using `StandardScaler` from `sklearn.preprocessing`, ensuring all variables were on a consistent scale with mean = 0 and standard deviation = 1. This helped prevent dominant features from biasing the model.

2.2 TEMPORAL ANALYSIS

Crash data was grouped as well as analyzed by time dimensions such as year, month, day of the week, and hour of the day. Annual trends revealed just how crash volumes shifted in course and included specific dips during 2020 because of the COVID-19 pandemic. Seasonal and hourly patterns pinpointed certain increases in crashes during the winter months. These increases were connected to road conditions and volumes of traffic during rush hours. This very comprehension of the temporal informed the conclusions of EDA and design of visualization.

2.3 GEOSPATIAL ANALYSIS

Spatial analysis was performed by plotting crash frequency across geographic units (latitude/longitude and county/city boundaries) using heatmaps and choropleths. Spatial clusters indicating high crash activity—particularly in urban areas such as Des Moines and Cedar Rapids—were revealed, and these were contrasted with rural crash patterns. This regional differentiation became central to the mapping dashboards and proved essential for predictive targeting.

2.4 CONTRIBUTING FACTORS

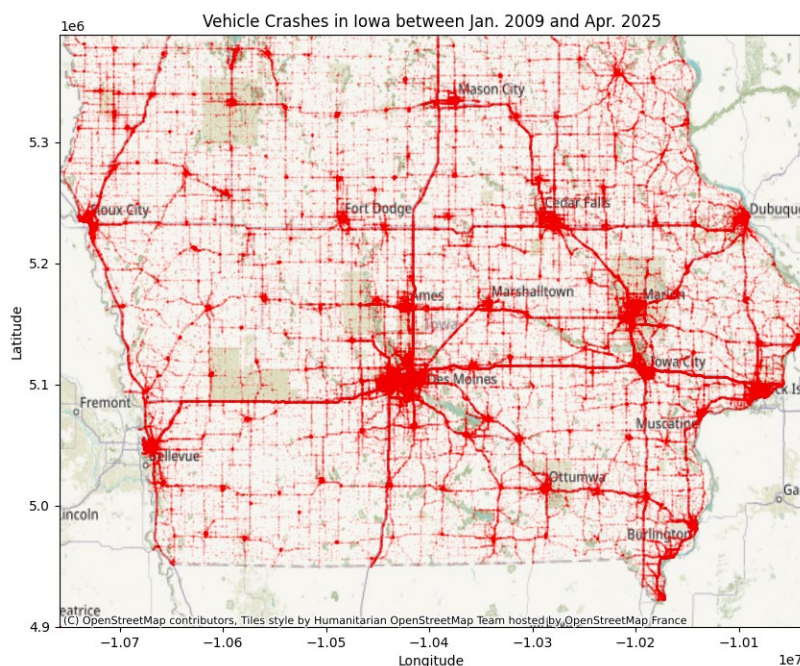
Explored key contributing variables, such as weather conditions and its influence on crash rates; road surface and lighting, such as the impact of wet surfaces and poor lighting on crash severity; and vehicle types and behavior, such as distribution of crashes by vehicle types and associated behaviors.

2.5 CRASH SEVERITY DISTRIBUTION

Graphed distribution of crashes by severity (i.e., property damage only, injury, fatality), which became the target variables in the predictive model developed. Further plots examined how severity related to variables such as time of day, weather, and driver demographics.

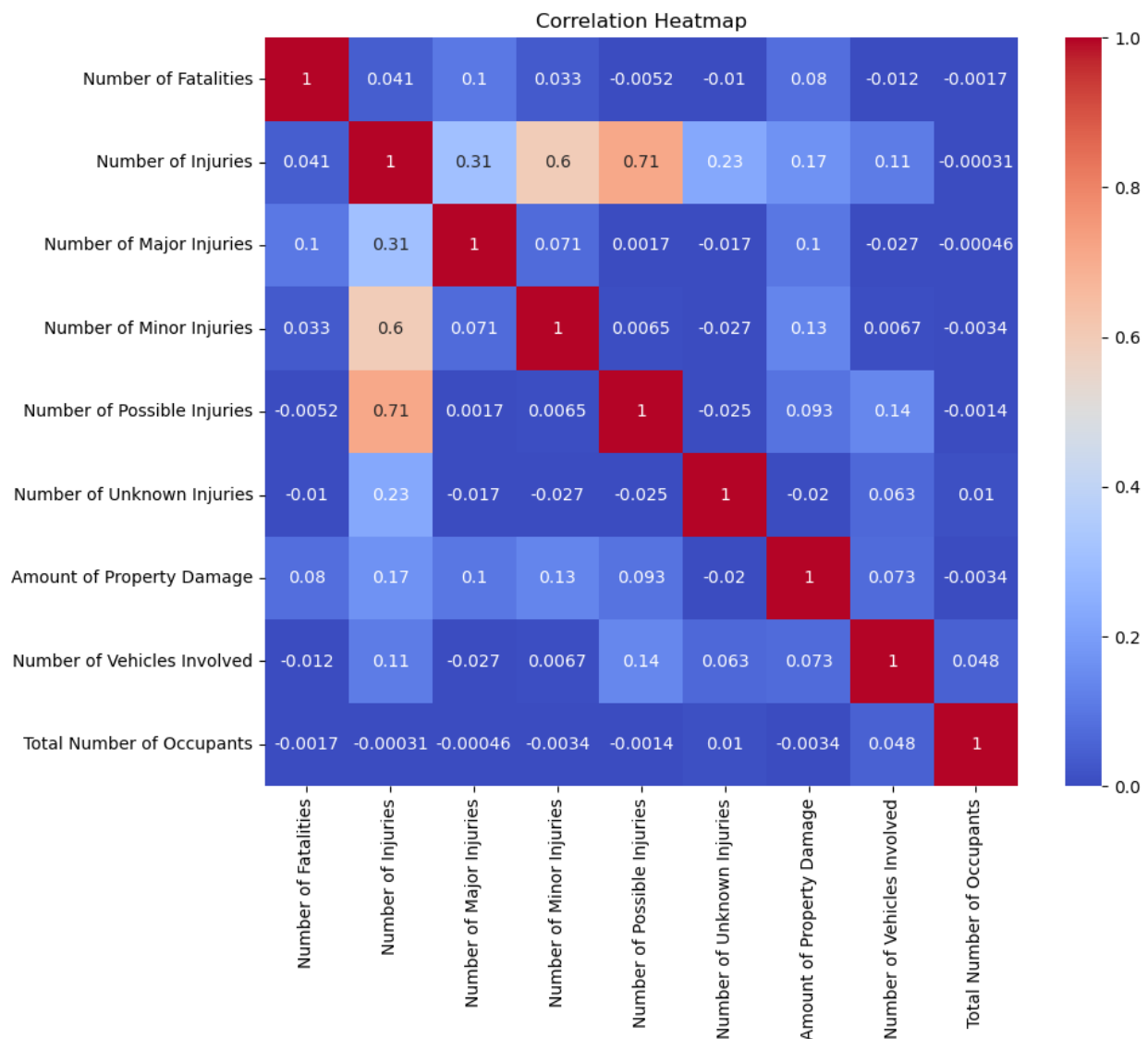
2.6 DOT-DENSITY MAPPING

With the help of geopandas, shapely, pyplot, and contextily (see Appendix B), a static dot-density map (pictured below) was produced which visualized vehicle crashes in the state of Iowa between January 2009 and April 2025. This revealed a greater frequency of crashes in major metropolitan areas, to be expected given the greater population—and number of vehicles otherwise driving through them—found within them.

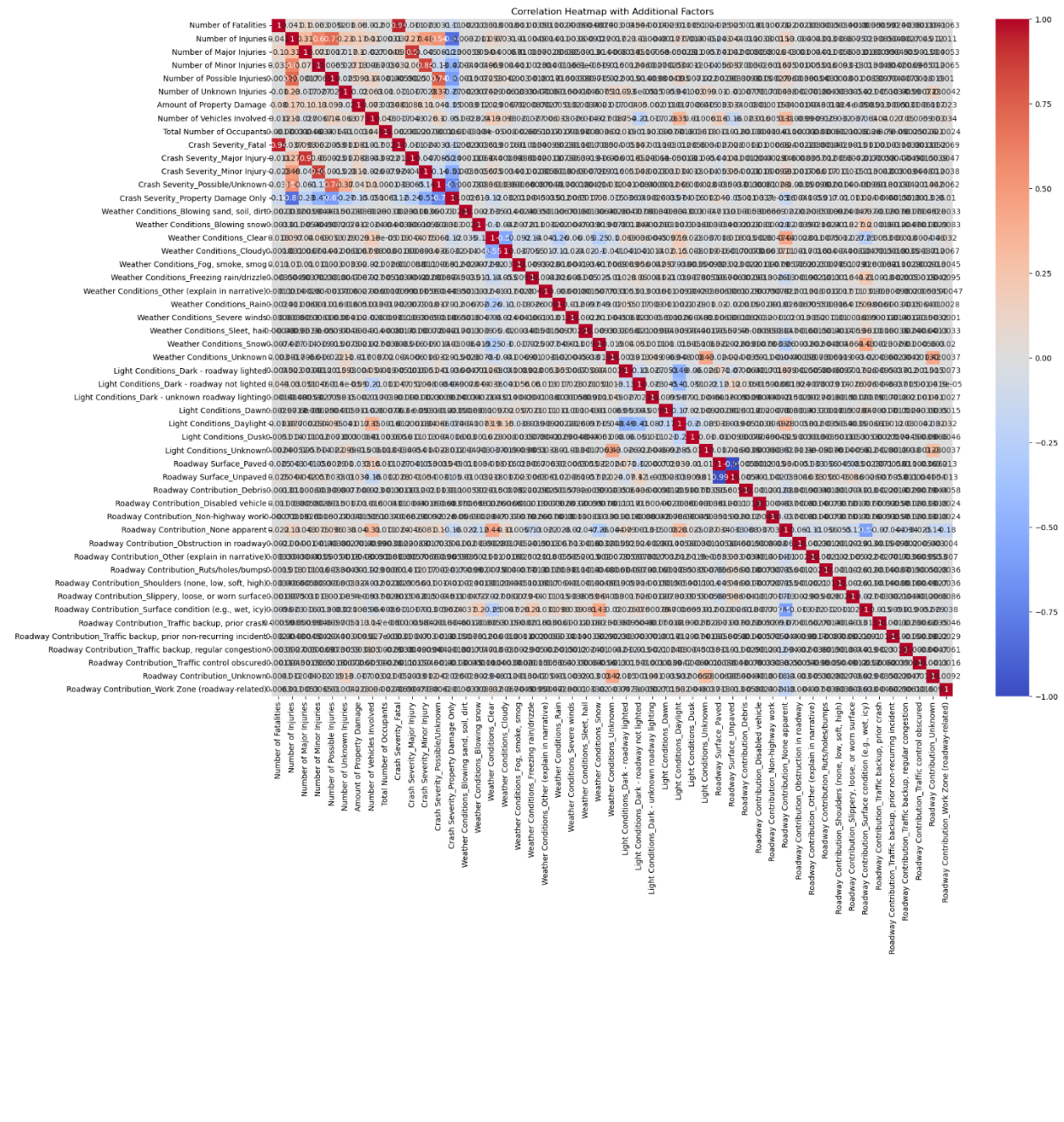


2.7 HEATMAPPING IMPORTANT FEATURES

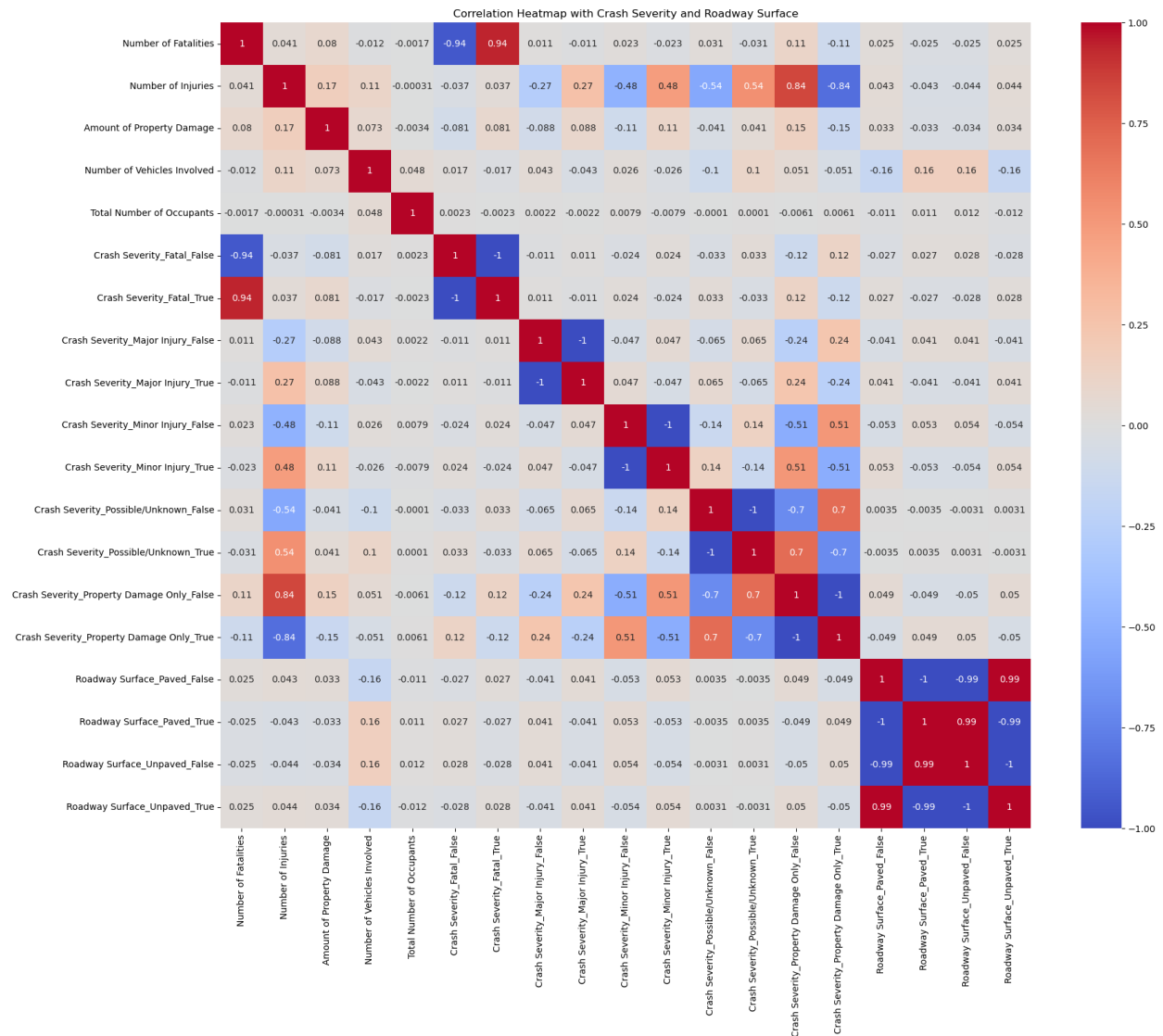
An exploratory analysis of vehicle crash data in Iowa was conducted, with a focus on uncovering relationships between different variables using heatmaps. Essential libraries such as pandas, seaborn, and matplotlib were imported for data manipulation and visualization. After the full dataset was loaded from a CSV file, a smaller sample comprising 5% of the data was created for quicker analysis and saved as a new file. The initial analysis was directed toward numeric variables, including the number of fatalities, injuries, property damage, and vehicles involved. A correlation matrix among these features was computed, and the results were visualized using a heatmap with a "coolwarm" color palette to help identify strong positive or negative relationships.



The analysis was expanded by incorporating several categorical variables, including crash severity, weather conditions, light conditions, roadway surface, and roadway contribution. These variables were transformed into numeric form through one-hot encoding. The correlation matrix was then recomputed to include both the original numerical features and the encoded categorical features. A larger heatmap was generated to provide a comprehensive view of how these factors interrelate.



Additional one-hot encoding was performed on the "Manner of Crash/Collision" and "Environmental Conditions" columns. The feature set was updated, the correlation matrix was recalculated, and the results were visualized using an even larger and more detailed heatmap. This final heatmap enabled deeper insight into how various crash circumstances and environmental factors may correlate with crash severity and outcomes.



2.8 ANALYSIS IN TABLEAU

To present key understandings derived from the Iowa vehicle crash dataset in a more intuitive and interactive format, 45 visualization charts were created using Tableau, enabling rapid insight generation and pattern identification. A series of dashboards were also developed using Tableau, with each dashboard focusing on a distinct aspect of crash-related data, including environmental factors, time patterns, causes, severity, demographics, and geographic distribution. As a whole, these dashboards were designed to provide stakeholders with an easy and comprehensive way to view the data and support data-driven decision-making.

2.8.1 Overall Crash Overview

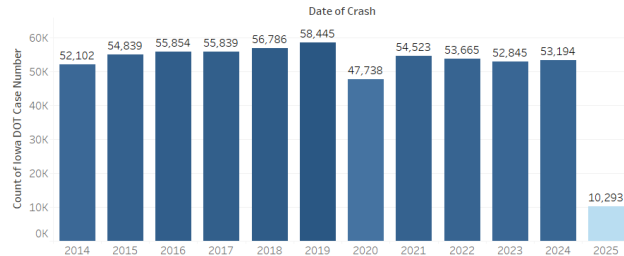
Total number of crashes

Total number of amount of property damage

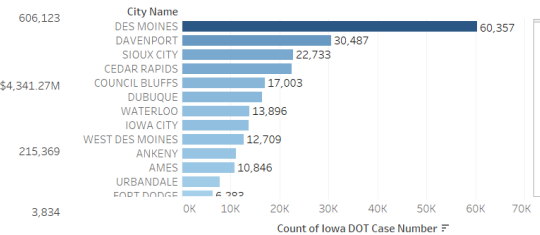
Total number of injuries

Total number of fatalities

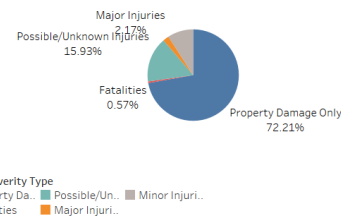
Crashes by year



Crashes by city in Iowa



Crashes by crash severity

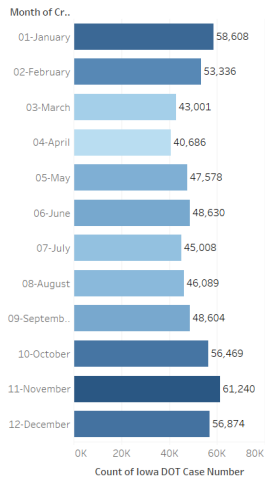


Dashboard 1 summarizes crash volume, severity, property damage, as well as geographic patterns in Iowa (2009–2025). A total of 606,123 crashes resulted in over \$4.34 billions in property damage, 215,369 injuries, and 3,834 fatalities. Des Moines reported the highest number of crashes (60,357), after Davenport and Sioux City. Most crashes (72.21%) involved only property damage, while fatalities accounted for just 0.57%.

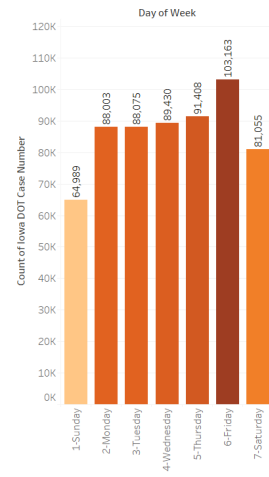
Highlight: Urban areas have high crash concentration; most cases are property damage only.

2.8.2 Crash Time Analysis

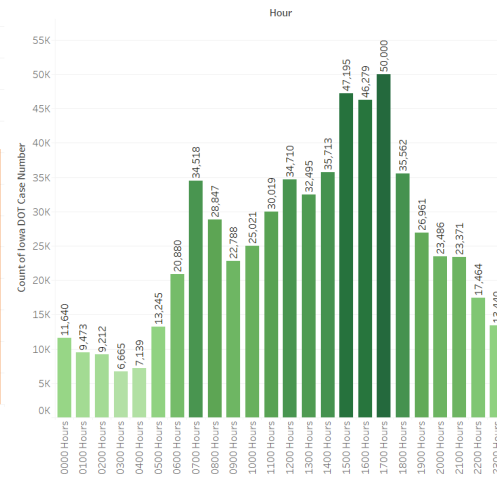
crashes by month of year



crashes by date of a week



crashes by hour of day

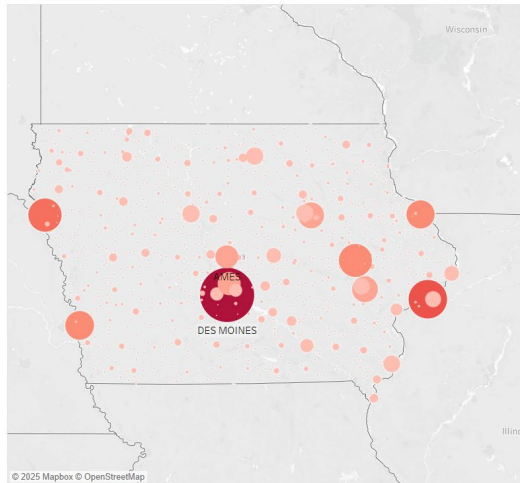


Dashboard 2 shows the times when crashes happen. Crashes peak in the winter months (Jan, Nov, Dec), most notably on Fridays, and in rush hours (8 AM and 4 PM). Crash rates are virtually lowest between 4–5 AM. These patterns do reflect the influence of the weather, in addition to traffic volume. Such patterns reflect the influence of weather as well as traffic volume.

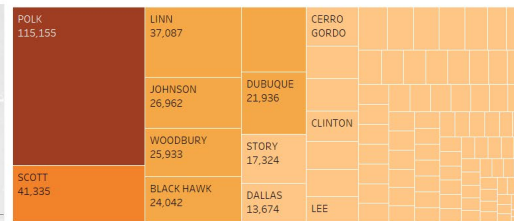
Highlight: Peak crash times are during morning and evening rush hours on weekdays.

2.8.3 Crash Geographic Distribution Analysis

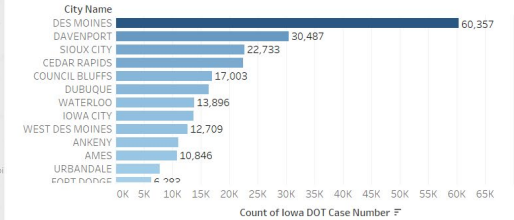
Crashes by city in Iowa-Map



Crashes by county in Iowa-TreeMap



Crashes by city in Iowa

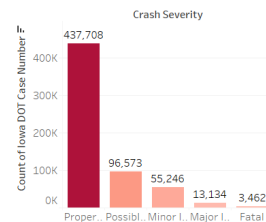


Dashboard 3 shows the spatial distribution of crashes across cities and counties. Des Moines as well as Polk County had the most incidents. Additional high-crash cities include Davenport, Sioux City, and Cedar Rapids. The map stresses urban areas with a higher population. The tree map stresses those with higher traffic density.

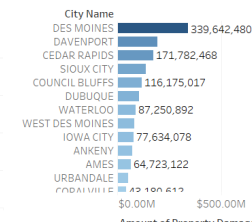
Highlight: Crash hotspots are mostly concentrated in Iowa's urban centers.

2.8.4 Crash Severity Analysis

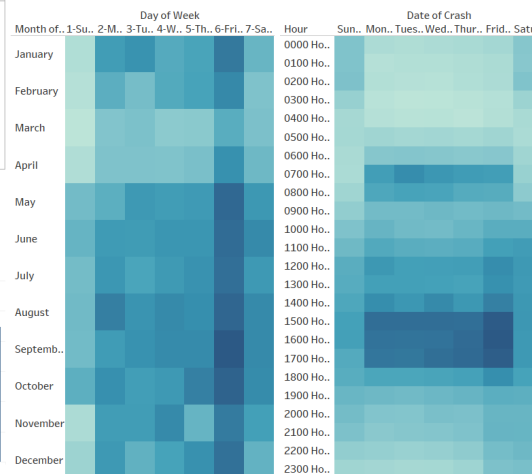
Crash severity distribution



Property damage by city

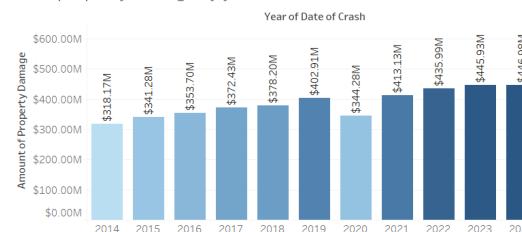


Distribution of Injuries by month and weekday



Distribution of injuries by weekday and hour

Total property damage by year

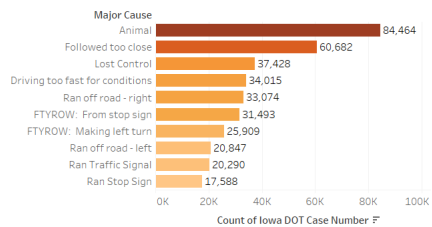


Dashboard 4 looks at the crash severity, injuries, and damage. Most of the crashes involved non-injury events (437,708), but there still were 3,462 fatalities. Des Moines leads in property damage, totaling \$339M. Total annual property damage increased steadily, peaking over at \$445M in 2023. Heatmaps show injury peaks throughout May–August and during normal weekday rush hours.

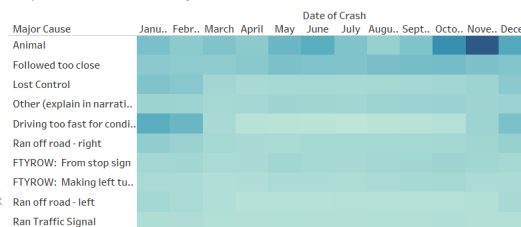
Highlight: Property damage as well as injury rates are closely tied to urban areas; peak travel times have similar trends.

2.8.5 Crash Major Causes Analysis

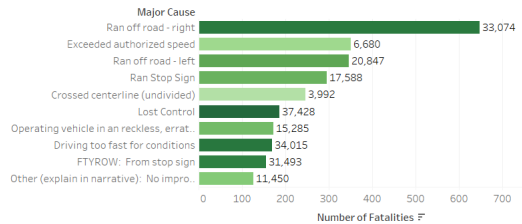
Top 10 Major Causes



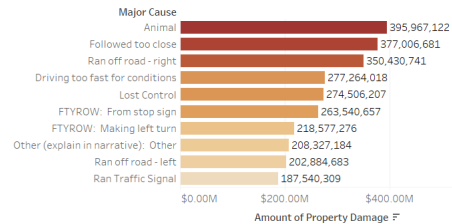
Top 10 Crash Causes by Month



Top 10 Contributing Causes of Fatal Vehicle Crashes



Top 10 Contributing Causes of property damage in Crashes

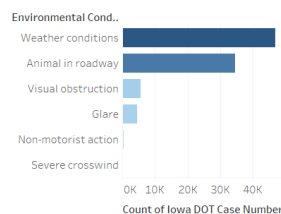


Dashboard 5 identifies the top causes. “Animal” and “Following too closely” comprise the most frequent crash causes. Fatal crashes were mostly due to speeding as well as vehicles running off the road. Crashes involving animals led to \$396M in damages. Speeding as well as reckless driving caused more than \$200M each.

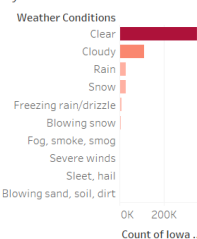
Highlight: Environmental and behavioral causes drive crash outcomes. These several causes include animal presence and speeding or tailgating.

2.8.6 Crash Environmental Conditions Analysis

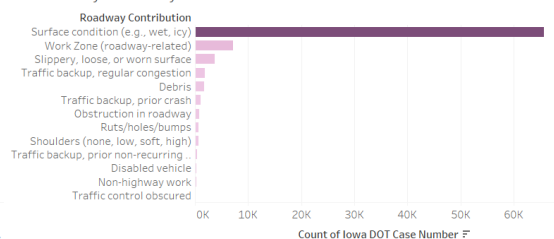
Most common environmental conditions



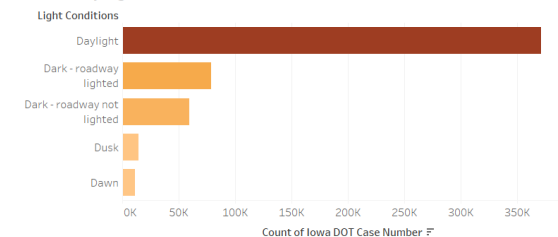
Crashes by weather conditions



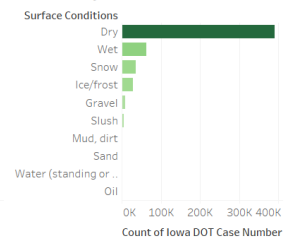
Crashes by roadway contribution



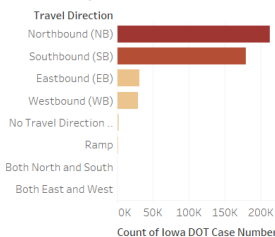
Crashes by light conditions



Crashes by surface conditions



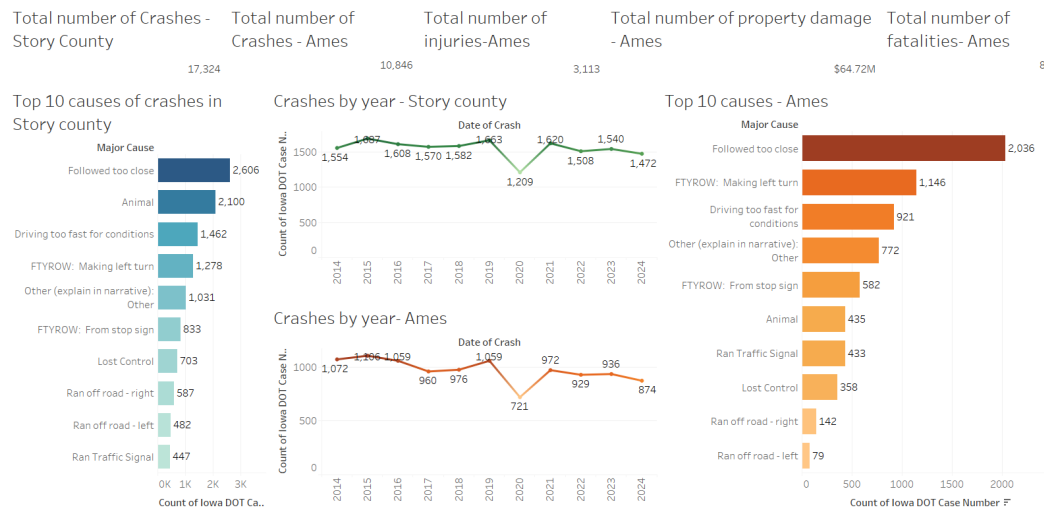
Crashes by travel direction



Dashboard 6 studies environmental and road conditions. Crashes happened mostly during clear weather, but crashes in rain/snow were of more severity. Poor road surfaces, in addition to dark, unlighted conditions, increased severity. Dry roads saw the most crashes, although winter weather created greater danger.

Highlight: Weather, the lighting, and the road surface do considerably affect crash severity.

2.8.7 Crash in Story County and Ames



Dashboard 7 is zooming into Story County and Ames. Ames had nearly 10,846 crashes of the county's nearly 17,324. In Ames, 3,113 injuries were reported, in addition to \$64.72M in damages. These totals were reported in Ames. "Following at a distance too closely" was, for both areas, the top cause. Crash counts dipped in 2020 due to COVID-19, although rebounded thereafter.

Highlight: Ames accounts for a meaningful proportion of Story County's crashes; causes mirror statewide patterns.

2.9 MODELING APPROACH

See Appendix A for code.

2.9.1 Feature Selection

Feature selection was conducted by explicitly selecting crash-descriptor columns, such as month, day of week, location, environmental conditions, cause, etc., as well as constructing new interaction features to capture potential nonlinear processes. For the property-damage target, capping at the 5th and 95th percentiles prevented extreme values from skewing the predictive power of the model, though at the cost of losing any potential for predictive ability for extreme outliers.

2.9.2 Model Selection

The robust and resource-efficient regressor model HistGradientBoostingRegressor was ultimately selected as the core algorithm for casualty and damage modeling, wrapped together in a MultiOutputRegressor. Specifically, it aided in the capture of the multifaceted interactions between many crash conditions and the target variables by binning the data for analysis rather than continuously analyzing the >600,000 data points.

2.9.3 Model Training

A training-testing split strategy was employed at an 80:20 train-test split with added reservation of the training data for additional automated validation in the event of overfitting, applied for each of the injury- and fatality-based targets.

2.9.4 Performance Evaluation

Metrics used included the computation of root mean square error, mean absolute error, and R-squared for each target, aggregated into a simple tabular format seen below. This allowed for comparison of the approach's efficacy across target variables

2.9.5 Hyperparameter Tuning

Setting the maximum number of iterations to 100 for the property damage target variable ensured that the most optimized solution was converged upon within the bounds specified for computational resource purposes. Further, this enables future application of GridSearchCV or RandomizedSearchCV given further exploration into model optimization for this context.

2.10 FINDINGS AND DISCUSSION

Test-set results (see Table 1) show that the fatality model achieves an RMSE of 0.0829, an MAE of 0.0113, and explains 11.2 % of the variance ($R^2 = 0.1120$). The injury model attains an RMSE of 0.6178, an MAE of 0.4363, and an R^2 of 0.1717. The property-damage model, trained on a log-transformed and percentile-capped target with median (quantile 0.5) loss, yields an RMSE of \$4673, an MAE of \$3176, and an R^2 of 0.1257. These results demonstrate that the pipeline predicts zero-inflated casualty counts with very low absolute error but captures only 11–17 % of their variability, while the cost model explains about 12.6 % of capped damage variance. In real-world terms, the casualty forecasts support reliable safety reporting, whereas the damage estimates—though improved by capping and median loss—remain best suited for high-level budgeting rather than precise financial planning.

Table 1: Metrics table results

target	dataset	RMSE	MAE	R2
Number of Fatalities	test	0.08291511	0.011302397	0.112032897
Number of Injuries	test	0.617787446	0.436333816	0.171724775
Amount of Property Damage	test	4673.186172	3176.222474	0.12573882

2.11 CONCLUSION AND RECOMMENDATIONS

Within this report, over 45 visual charts across 7 dashboards were developed, illustrating crash frequency, severity, timing, causes, demographics, and geographic patterns. Tableau was used to explore complex data trends and to communicate these patterns in a clear, accessible format. The dashboards revealed key findings, including a high concentration of crashes in urban areas such as Des Moines, seasonal and weekday peaks in crash occurrences, and a predominance of property-damage-only incidents. These insights demonstrated the value of data-driven tools in shaping traffic safety strategies.

Based on the insights drawn from the Tableau dashboards, it is recommended that transportation planners and safety authorities prioritize interventions in high-crash areas and during high-risk time periods. For instance, enhanced winter road maintenance, improved lighting in poorly lit zones, and upgraded traffic controls in cities like Des Moines could help reduce crash severity. Law enforcement agencies are advised to target specific behavioral factors—such as speeding and following too closely—through focused enforcement during peak hours. Additionally, insurance

companies and policymakers are encouraged to use these tools to identify regional risk profiles and allocate resources more effectively.

Although the project provided valuable visual insights through Tableau, there are several limitations. The dataset contained missing values in key fields, such as travel direction and work zone status, and injury categories were broadly defined. While Tableau excelled in data exploration and communication, it lacked the capacity for advanced predictive modeling, such as machine learning.

Moving ahead, the analysis set out to build predictive models for three critical outcomes of Iowa vehicle crashes—fatalities, injuries, and property damage costs—using a consistent pipeline of feature engineering, gradient boosted regression, and robust evaluation. It ingested 21 primary crash descriptors plus five interaction terms, applied percentile-based capping and log transformation to stabilize extreme cost values, and trained separate HistGradientBoostingRegressor pipelines for casualty counts and median quantile damage. On held out data, the fatality model produced an RMSE of 0.083 deaths and explained 11.2 % of variance; the injury model yielded an RMSE of 0.618 injuries and R^2 of 17.2 %; and the damage model achieved an RMSE of \$4,673, MAE of \$3,176, and R^2 of 12.6 % on capped costs. These results demonstrate consistently low absolute errors for rare-event counts and moderate explanatory power for cost predictions.

Stakeholders can leverage these casualty models to forecast overall safety trends and allocate resources toward high-risk corridors, knowing that per-crash fatality and injury predictions carry subunit error. For budgeting or insurance reserve planning, the property damage estimates—while improved by outlier capping and median-focused loss—are best suited to scenario-level cost projections rather than granular claim-level accuracy. To enhance operational value, stakeholders should integrate the models into routine crash reporting dashboards, combine them with GIS-based hot spot analysis, and use predicted injury counts to inform emergency response staging.

The study's limitations include the relatively low R^2 across all targets, a reflection of zero inflated casualty data and heavy skew in cost distributions. Capping at the 5th/95th percentiles trades off extreme-event fidelity for overall fit, and ordinal encoding may underutilize high cardinality categorical nuances. The current pipelines also omit temporal cross validation, spatial autocorrelation, and richer external covariates (e.g. traffic volume, vehicle age, roadway speed limits).

Future work should explore specialized count- and cost-model families (e.g., Poisson, Tweedie, Gamma regressors), advanced target encoding schemes for city or route effects, and potentially hyperparameter optimization via advanced Bayesian search. Incorporating real time weather feeds, temporal rolling windows, and more granular location data (e.g., road segment identifiers) would also further strengthen predictive accuracy and deepen actionable insights.

2.12 REFERENCES

Iowa Department of Transportation, Motor Vehicle Division. "Vehicle Crashes in Iowa." *Data.Iowa.Gov*, 25 Mar. 2025, https://data.iowa.gov/Crashes/Vehicle-Crashes-in-Iowa/tw78-ziwj/about_data.

2.13 APPENDICES

2.13.1 Appendix A

```
df = pd.read_excel('Vehicle_Crashes_in_Iowa_20250410.xlsx')

# Capping property damage at 1st and 99th percentiles to prevent skewing
low, high = df['Amount of Property Damage'].quantile([0.05, 0.95])
df['Damage_capped'] = df['Amount of Property Damage'].clip(low, high)

# Log-transforming the capped damage for analysis
df['Damage_log'] = np.log1p(df['Damage_capped'])

# Creating feature interactions
df['Light_x_Weather'] = df['Light Conditions'] + ':' + df['Weather Conditions']
df['MajorCause_x_Surface'] = df['Major Cause'] + ':' + df['Surface Conditions']
df['Route_x_Event'] = df['Route with System'] + ':' + df['First Harmful Event']
df['WorkZone_x_District'] = df['Work Zone'] + ':' + df['DOT District']
df['Month_x_Day'] = df['Month of Crash'] + ':' + df['Day of Week']

# Defining the features and targets for analysis
features = [
    'Month of Crash', 'Day of Week', 'Hour', 'DOT District', 'City Name', 'County Name',
    'Route with System', 'First Harmful Event', 'Location of First Harmful Event',
    'Manner of Crash/Collision', 'Major Cause', 'Drug or Alcohol ', 'Environmental Conditions',
    'Light Conditions', 'Surface Conditions', 'Weather Conditions', 'Roadway Contribution',
    'Roadway Type', 'Roadway Surface', 'Work Zone', 'Total Number of Occupants',
    'Light_x_Weather', 'MajorCause_x_Surface', 'Route_x_Event',
    'WorkZone_x_District', 'Month_x_Day'
]
target_casualties = ['Number of Fatalities', 'Number of Injuries']
target_damage_log = 'Damage_log'
target_damage_orig = 'Damage_capped'

# Converting categorical columns to string to resolve type issues with OneHotEncoder
for col in categorical_features:
    df[col] = df[col].astype(str)

numeric_features = ['Total Number of Occupants']
categorical_features = [col for col in features if col not in numeric_features]

# Preparing training/testing data split
X = df[features]
y_cas = df[target_casualties]
y_dmg_log = df[target_damage_log]
y_dmg_orig = df[target_damage_orig]

X_train, X_test, y_train_cas, y_test_cas, y_train_dmg_log, y_test_dmg_log, y_train_dmg_orig,
y_test_dmg_orig = train_test_split(
    X, y_cas, y_dmg_log, y_dmg_orig, test_size=0.2, random_state=42
)

# Preprocessing pipeline
preprocessor = ColumnTransformer([
    ('num', SimpleImputer(strategy='median'), numeric_features),
    ('cat', Pipeline([
        ('impute', SimpleImputer(strategy='constant', fill_value='Missing')),
        ('ord', OrdinalEncoder(handle_unknown='use_encoded_value', unknown_value=-1))
    ]), categorical_features)
])

# Casualties pipeline
hgb_cas = HistGradientBoostingRegressor(random_state=42, max_iter=100, early_stopping='auto')
```

```

cas_pipeline = Pipeline([
    ('pre', preprocessor),
    ('model', MultiOutputRegressor(hgb_cas))
])
cas_pipeline.fit(X_train, y_train_cas)
y_test_cas_pred = cas_pipeline.predict(X_test)

# Damage pipeline
hgb_dmg = HistGradientBoostingRegressor(
    loss='quantile',
    quantile=0.5,
    random_state=42,
    max_iter=100,
    early_stopping='auto'
)
dmg_pipeline = Pipeline([
    ('pre', preprocessor),
    ('model', hgb_dmg)
])
dmg_pipeline.fit(X_train, y_train_dmg_log)
y_test_dmg_log_pred = dmg_pipeline.predict(X_test)

# Back-transforming predictions to original scale
y_test_dmg_pred = np.expml(y_test_dmg_log_pred)

# Evaluating and collecting metrics
rows = []

# Casualties metrics
for idx, target in enumerate(target_casualties):
    rows.append({
        'target': target,
        'dataset': 'test',
        'RMSE': root_mean_squared_error(y_test_cas[target], y_test_cas_pred[:, idx]),
        'MAE': mean_absolute_error(y_test_cas[target], y_test_cas_pred[:, idx]),
        'R2': r2_score(y_test_cas[target], y_test_cas_pred[:, idx])
    })

# Damage metrics (original scale)
rows.append({
    'target': 'Amount of Property Damage',
    'dataset': 'test',
    'RMSE': root_mean_squared_error(y_test_dmg_orig, y_test_dmg_pred),
    'MAE': mean_absolute_error(y_test_dmg_orig, y_test_dmg_pred),
    'R2': r2_score(y_test_dmg_orig, y_test_dmg_pred)
})

metrics_df = pd.DataFrame(rows)
display(metrics_df)

```

2.13.2 Appendix B

```
import time
import pandas as pd
import geopandas as gpd
from shapely import wkt
from tqdm import tqdm
import matplotlib.pyplot as plt
import contextily as ctx

# Making my dataframe from the data
df = pd.read_excel('Vehicle_Crashes_in_Iowa_20250410.xlsx')

# Removing blank rows
df = df[df['Location'].notna()]
df = df[df['Location'].str.strip() != ""]

# Converting *valid* WKT strings into geometry objects
df['geometry'] = df['Location'].apply(wkt.loads)

# Creating a GeoDataFrame
gdf = gpd.GeoDataFrame(df, geometry='geometry')
gdf.crs = "EPSG:4326" # WGS84 (latitude and longitude)

# Dropping any points where x and y are close to zero
tol = 1 # tolerance threshold
gdf = gdf[~((gdf.geometry.x.abs() < tol) & (gdf.geometry.y.abs() < tol))]

# Converting the CRS to EPSG:3857 to overlay on a basemap (Contextily requirement)
gdf = gdf.to_crs(epsg=3857)

# Creating a Matplotlib figure and axis
fig, ax = plt.subplots(figsize=(10, 8))
ax.set_title("Vehicle Crashes in Iowa between Jan. 2009 and Apr. 2025")
ax.set_xlabel("Longitude")
ax.set_ylabel("Latitude")

# Setting up a base map
xmin, ymin, xmax, ymax = gdf.total_bounds
ax.set_xlim(xmin - 1, xmax + 1)
ax.set_ylim(ymin - 1, ymax + 1)

# Adding markers to the plot (also with a progress bar)
for idx, row in tqdm(gdf.iterrows(), total=gdf.shape[0], desc="Plotting markers"):
    ax.plot(row.geometry.x, row.geometry.y, marker='.', color='red', markersize=1, alpha=0.1)

# Underlaying the base map
ctx.add_basemap(ax, crs=gdf.crs)

# Saving the static map to an HTML file
# Also, measuring the time it takes to save the static map as an image
start_time = time.time()
plt.savefig("static_map.png")
end_time = time.time()
print(f"plt.savefig() took {end_time - start_time:.2f} seconds")
print("Static map saved as 'static_map.png'.")
# plt.show()
```